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Predicting Consumer Sentiment: A Multi-class NLP Pipeline for the Amazon Fine Food Review Dataset

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1. Motivation

The rapid expansion of e-commerce platforms has led to an overwhelming volume of customer reviews that significantly influence purchasing decisions. Manually analysing such large-scale textual data is inefficient, creating a need for automated approaches that can transform unstructured reviews into actionable insights. The Amazon Fine Food Reviews dataset [1] exemplifies this challenge, as it contains extensive user feedback that reflects varying levels of customer satisfaction. The motivation for this project is to utilize machine learning techniques to automatically predict review scores based on textual content. Accurate rating prediction helps businesses quickly identify dissatisfied customers, monitor satisfaction trends, and support data-driven decision-making. Additionally, customer reviews often include nuanced language and mixed sentiments, making this task well-suited for Natural Language Processing (NLP) methods. This project formulates the problem as a supervised multi-class classification task aimed at predicting ratings on a scale from 1 to 5. The model uses two primary input features — the review Summary and Text — which are preprocessed and converted into numerical representations for training. The output is a categorical score representing the user-assigned rating. Figure 1 presents the overall workflow of the proposed system, including preprocessing, model training, and prediction.

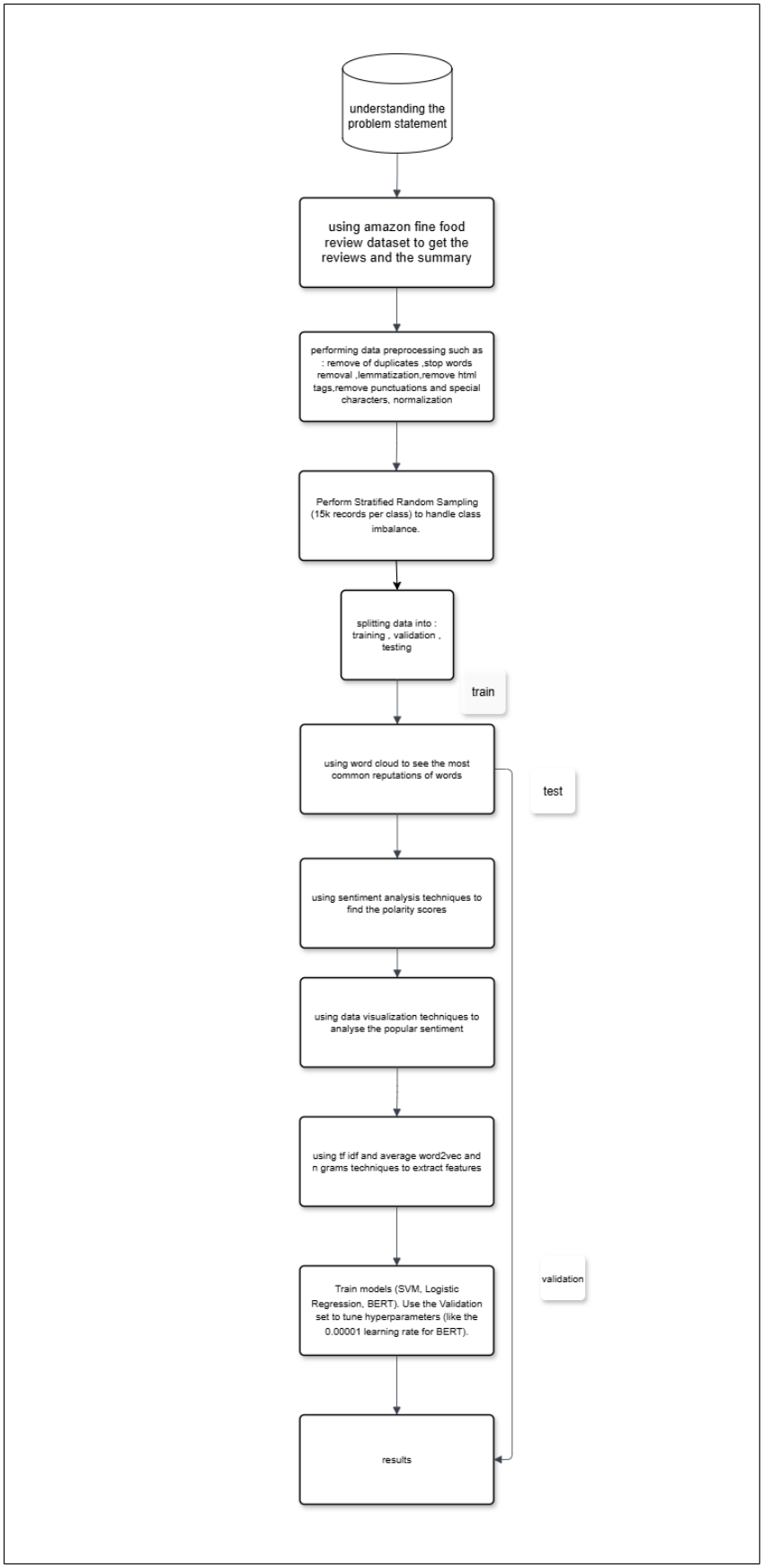

Figure 1:Proposed System Architecture for Multi-class Sentiment Classification

2 Background

2.1 Key Concepts and Background Information

Natural Language Processing (NLP): Natural Language Processing enables machines to understand and analyze human language. In this project, NLP techniques are applied to convert textual reviews into structured data that can be used for machine learning models [2].

Supervised Multi-class Classification: Multi-class classification is a supervised learning task where each input is assigned to one of more than two categories. Unlike binary sentiment classification (positive vs. negative), this project addresses a fine-grained 5-class problem [3].

Sentiment Analysis (Fine-Grained): Sentiment Analysis refers to the automatic identification of opinions, emotions, or attitudes expressed in text. In the context of this study, sentiment analysis is used to predict a customer's rating score (1–5 stars) based solely on the written review content [4].

Class Imbalance: Online reviews typically follow a "J-shaped" distribution, where extreme ratings dominate. This imbalance can affect model performance and must be considered during model development [5].

2.2 Summary of Related Works

Online review analysis has been widely studied in natural language processing and machine learning, particularly using the Amazon Fine Food Reviews dataset. Prior research has explored both multi-class rating prediction and binary sentiment classification, employing traditional machine learning and deep learning approaches.

One of the most directly related works to our task is the study by Zhao and Sun (2022) [6], which focuses on predicting the 1–5 star rating from review text using a fine-tuned BERT model . The authors applied extensive preprocessing, including duplicate removal, language filtering, punctuation cleaning, and class rebalancing. They fine-tuned a pre-trained *bert-base-cased* transformer model for 5-class classification and achieved a test accuracy of 79.82%. Their results show that deep contextual embeddings can capture semantic nuances in review text; however, the performance was affected by class imbalance and the inherent subjectivity of ratings. This work is highly relevant to our project since we aim to solve the same multi-class score prediction problem.

In contrast, Das et al. (2020) [7] approached the problem as a binary sentiment classification task, converting ratings into positive (4–5 stars) and negative (1–2 stars) classes . They compared multiple feature extraction methods such as Bag-of-Words (BoW), TF-IDF, and Average Word2Vec embeddings, combined with Logistic Regression and Support Vector Machines (SVM). Their best-performing model (Linear SVM with Average Word2Vec) achieved 93.11% accuracy on the test set. The study demonstrates that well-engineered features combined with linear classifiers can outperform more complex models for binary sentiment tasks. However, since their formulation simplifies the problem into two classes, it does not directly address fine-grained rating prediction.

Similarly, Thakkar et al. (2020) [8] conducted a comparative analysis of several machine learning and deep learning models, including Logistic Regression, Naïve Bayes, Decision Trees, Random Forests, and LSTM networks . Their experiments showed that Logistic Regression and Naïve Bayes achieved competitive results, while LSTM reached approximately 92% accuracy in binary classification. The authors emphasized the importance of preprocessing techniques such as stop-word removal, stemming, and feature engineering (bi-grams and TF-IDF). Their findings reinforce that model performance heavily depends on feature representation and data preparation rather than solely on model complexity.

Another deep learning–based approach was proposed by Wu and Ji, who applied Recursive Neural Networks (RNN) to sentiment analysis on the Amazon Fine Food Reviews dataset [1]. Unlike traditional bag-of-words or flat neural models, their method incorporates syntactic structure by constructing binary parse trees using the Stanford NLP Parser. Since the dataset provides only one label per review, the authors introduced a technique to partially label internal tree nodes to increase supervision during training. They also proposed an extended model, RNNMS (Recursive Neural Network for Multiple Sentences), designed to aggregate sentence-level representations at the review level. Their model achieved an average F1-score of 0.74, outperforming a Naïve Bayes baseline that achieved 0.69. However, the study highlighted that recursive neural networks require strong supervision and sufficient labeling to reach higher accuracy, which can be challenging in real-world review datasets.

Overall, prior studies reveal that multi-class score prediction is more challenging than binary sentiment classification due to finer-grained rating distinctions. Deep learning approaches such as BERT and Recursive Neural Networks demonstrate the ability to capture contextual and structural information in review text but often require careful preprocessing, sufficient supervision, and class balancing. At the same time, traditional machine learning models combined with effective feature engineering techniques (e.g., TF-IDF and Word2Vec) remain

highly competitive, particularly for binary sentiment tasks. These findings highlight the trade-offs between model complexity, computational cost, and predictive performance in review-based rating prediction problems.

3 Dataset

3.1 Overview and Source

the Amazon Fine Food Review dataset [1] is a comprehensive collection of consumer feedback spanning over a decade (October 1999 – October 2012). This dataset was originally compiled and analysed by researchers at Stanford University to study the evolution of user expertise in online review systems [9]. It captures the multifaceted nature of product reviews, including user identity, product metadata, and high-granularity textual feedback.

3.2 Dataset Rationale

This dataset was selected due to its alignment with the problem of multi-class sentiment intensity. Unlike smaller datasets, the 568,454 reviews provide enough statistical variance to train complex models without overfitting. Furthermore, the inclusion of both Summary and Text fields allows the model to learn from both concise and detailed linguistic patterns, which is essential for accurate star-rating prediction.

3.3 Statistical summary

feature	Value
Total Observations	568,454 Reviews
Unique Products	74,258
Unique Users	256,059
columns	Id, ProductId, UserId, ProfileName, HelpfulnessNumerator, HelpfulnessDenominator, Score, Time, Summary, Text
Number Of features	10
Users with > 50 reviews	260
Median no. of words per review	56
Timespan	Oct 1999 - Oct 2012

Table 1: summary statistics Dataset Table

3.4 Representative Examples

To illustrate the nature of the data, the following figure provides a snapshot of the raw records from the dataset, highlighting the relationship between the textual summary and the assigned numerical score.

Id	ProductId	UserId	ProfileNan	Helpfulnes	Helpfulnes	Score	Time	Summary	Text
1	B001E4KF	A3SGXH7A	delmartian	1	1	5	1.3E+09	Good Qual	I have bought several of the Vitality canned dog food products and have found them all to be of good quality. The prod
2	B00813GR	A1D87F6Z	dll pa	0	0	1	1.35E+09	Not as Adv	Product arrived labeled as Jumbo Salted Peanuts...the peanuts were actually small sized unsalted. Not sure if this wa
3	B000LQOC	ABXLMWJI	Natalia Co	1	1	4	1.22E+09	"Delight" s	This is a confection that has been around a few centuries. It is a light, pillowy citrus gelatin with nuts - in this case Fil
4	B000UA0C	A395BORC	Karl	3	3	2	1.31E+09	Cough Me	If you are looking for the secret ingredient in Robitussin I believe I have found it. I got this in addition to the Root Beer
5	B006K2ZZ	A1UQRSCI	Michael D.	0	0	5	1.35E+09	Great taffy	Great taffy at a great price. There was a wide assortment of yummy taffy. Delivery was very quick. If your a taffy lover
6	B006K2ZZ	ADT0SRK1	Twoapenn	0	0	4	1.34E+09	Nice Taffy	I got a wild hair for taffy and ordered this five pound bag. The taffy was all very enjoyable with many flavors: watermelo
7	B006K2ZZ	A1SP2KVK	David C. S	0	0	5	1.34E+09	Great! Jus	This saltwater taffy had great flavors and was very soft and chewy. Each candy was individually wrapped well. None o
8	B006K2ZZ	A3JRGQVE	Pamela G.	0	0	5	1.34E+09	Wonderful	This taffy is so good. It is very soft and chewy. The flavors are amazing. I would definitely recommend you buying it. V
9	B000E7L2I	A1MZYO9	R. James	1	1	5	1.32E+09	Yay Barley	Right now I'm mostly just sprouting this so my cats can eat the grass. They love it. I rotate it around with Wheatgrass a
10	B00171AP	A21BT40V	Carol A. Re	0	0	5	1.35E+09	Healthy Do	This is a very healthy dog food. Good for their digestion. Also good for small puppies. My dog eats her required amoun
11	B0001PB9	A3HDKO7	Canadian I	1	1	5	1.11E+09	The Best H	I don't know if it's the cactus or the tequila or just the unique combination of ingredients, but the flavour of this hot sa
12	B0009XLV	A2725IB4Y	A Poeng "S	4	4	5	1.28E+09	My cats LC	One of my boys needed to lose some weight and the other didn't. I put this food on the floor for the chubby guy, and th
13	B0009XLV	A327PCT2	LT	1	1	1	1.34E+09	My Cats Ar	My cats have been happily eating Felidae Platinum for more than two years. I just got a new bag and the shape of the
14	B001GVISJ	A18ECVX2	willie "roac	2	2	4	1.29E+09	fresh and g	good flavor! these came securely packed... they were fresh and delicious! i love these Twizzlers!
15	B001GVISJ	A2MUGFV	Lynrie "Oh	4	5	5	1.27E+09	Strawberry	The Strawberry Twizzlers are my guilty pleasure - yummy. Six pounds will be around for a while with my son and I.
16	B001GVISJ	A1CZX3CF	Brian A. Le	4	5	5	1.26E+09	Lots of twi	My daughter loves twizzlers and this shipment of six pounds really hit the spot. It's exactly what you would expect...si
17	B001GVISJ	A3KLWF6V	Erica Neat	0	0	2	1.35E+09	poor taste	I love eating them and they are good for watching TV and looking at movies! It is not too sweet. I like to transfer them t
18	B001GVISJ	AFKW14U	Becca	0	0	5	1.35E+09	Love it!	I am very satisfied with my Twizzler purchase. I shared these with others and we have all enjoyed them. I will definite
19	B001GVISJ	A2A9X58G	Wolfee1	0	0	5	1.32E+09	GREAT SW	Twizzlers, Strawberry my childhood favorite candy, made in Lancaster Pennsylvania by Y & S Candies, Inc. one of the
20	B001GVISJ	A3IV7CL2	Greg	0	0	5	1.32E+09	Home deli	Candy was delivered very fast and was purchased at a reasonable price. I was home bound and unable to get to a sto
21	B001GVISJ	A1WO0KG	mom2emr	0	0	5	1.31E+09	Always fre	My husband is a Twizzlers addict. We've bought these many times from Amazon because we're government employe
22	B001GVISJ	AZOF9E17	Tammy An	0	0	5	1.31E+09	TWIZZLERS	I bought these for my husband who is currently overseas. He loves these, and apparently his staff likes them also.<br
23	B001GVISJ	ARYVQL4	Charles Br	0	0	5	1.3E+09	Delicious p	I can remember buying this candy as a kid and the quality hasn't dropped in all these years. Still a superb product you

Figure 2:Sample of raw Amazon Fine Food Reviews.

4 Contributions

Name	Contribution
Aryam Almutairi	Responsible for drafting the Motivation section, including defining the research problem, explaining its significance, and clearly formulating the supervised learning task with specified input and output.
Leen binmueqal	Developed the Key Concepts, presenting the theoretical background on supervised learning, multi-class classification, and natural language processing techniques relevant to the task.
Nora Alkhudair	Prepared the Summary of Related Work by reviewing selected research papers and synthesizing prior approaches to rating prediction and sentiment analysis.
Malak Basloom	Completed the Dataset section by describing the data source, providing statistical summaries and representative examples.

Table 2:Project Contribution Table

5 References

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